Graduate course · For the social sciences

Advanced Data Analysis

Assignment Book — five exercises, from the first loop to language models as measurement instruments

Instructor	M. Fuat Kına
Contact	fuatkina@gmail.com · mkina@ku.edu.tr

Environment	Python (Anaconda / Google Colab / Jupyter / VS Code)
Submission	One GitHub repository named `Python_Course`, with the instructor and TA added as collaborators. Standard folder and file names: see the Submission Guide on the next page.

Contents

#	Assignment	Released	Core dataset / source	Points
1	Programming fundamentals	Week 3	Self-contained problems (no dataset)	8
2	Data wrangling with Pandas	Week 5	Gapminder World (course GitHub)	8
3	Web scraping and APIs	Week 6	A scrape-permitted website, or a platform API	8
4	Machine learning (shared task)	Week 9	SOMOS social-movement organisations (course GitHub)	13
5	LLM-based annotation	Week 11	Your own labelled corpus (ground truth)	8

General Rules

- Assignments are 45% of the final grade: Assignment 4 counts 13 points, the others 8 each.
- Discuss problems with classmates if you like, but submit **your own** code and writing.
- Readability counts: use Markdown cells and comments.
- Notebooks must run top to bottom without errors; record library versions and random seeds.
- Never commit credentials: keep API keys in a `.env` file listed in `.gitignore`.

Submission Guide

All five assignments go in **one** repository with the structure below. Nothing else is required.

```
Python_Course/
├── README.md                  # name, student ID, one link per assignment folder
├── assignment_1/
│   └── A1_notebook.ipynb
├── assignment_2/
│   └── A2_notebook.ipynb
├── assignment_3/
│   ├── A3_notebook.ipynb
│   ├── A3_data.csv            # collected data or a sample
│   └── A3_report.pdf
├── assignment_4/
│   ├── A4_notebook.ipynb
│   ├── A4_target_coding.csv   # ideology → right_wing mapping
│   └── A4_report.pdf          # one page
├── assignment_5/
│   ├── A5_notebook.ipynb
│   ├── A5_annotations.xlsx
│   └── A5_report.pdf          # one page
└── .gitignore                 # includes .env (API keys)
```

Conventions (the same for every assignment)

- **Notebook:** `A<n>_notebook.ipynb`, saved **with outputs visible**; the first cell gives your name, the assignment, and the date.
- **Report:** `A<n>_report.pdf` (Assignments 3–5).
- **Data:** only where requested, as `.csv` or `.xlsx`; no large raw files.
- **Deadline:** the last commit before the deadline counts.

Assignment 01 *8 points · Jupyter notebook*

Programming Fundamentals

Module I · Released Week 3

Objective	Solve five short programming problems.
Dataset	None.
Outputs	Folder `assignment_1/`: `A1_notebook.ipynb`, one section per problem, saved with outputs.

Problems

01 **Number guessing game.** Draw a random integer from 1 to 20 and let the user guess until correct, answering “higher” or “lower” after each wrong guess.

02 **Search insert position.** Given a sorted array of distinct integers and a target, return the target’s index, or the index where it would be inserted.

03 **Add binary strings.** Return the sum of two binary strings as a binary string (`"11"` + `"1"` → `"100"`).

04 **Single number.** In an array where every element appears twice except one, find that one.

05 **Repeated DNA sequences.** Return all 10-letter substrings that occur more than once in a DNA string (`"AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT"` → `["AAAAACCCCC", "CCCCCAAAAA"]`).

> ***Note***
>
> Plain code with clear reasoning earns more than elegant code without it.

Assignment 02 *8 points · Jupyter notebook*

Data Wrangling with Pandas

Module II · Released Week 5

Objective	Clean, merge, and explore a country–year dataset with Pandas.
Dataset	**Gapminder World**, on the course GitHub page in `Datasets/gapminder/`: `gdp_pcap.csv`, `life_expectancy.csv`, `population.csv`. Original source: gapminder.org/data. Continent information is not included; finding it is part of the task.
Outputs	Folder `assignment_2/`: `A2_notebook.ipynb`, one section per task, saved with outputs.

Tasks

01 **Load and merge.** Reshape the three files to long format and merge them into one country–year table. Add a continent classification from a source you cite, and report the countries that fail to match.

02 **Clean.** Handle missing values (justify your choice) and give the columns descriptive names.

03 **Explore.** Show the first and last rows, count the countries, and extract a single year.

04 **Filter and organise.** Select one continent, drop countries under one million people, sort by life expectancy, and classify population size with thresholds you explain.

05 **Analyse and visualise.** Compute average GDP per capita by continent, find the country with the highest life expectancy in a given year, and plot one country's life expectancy over time.

Assignment 03 *8 points · Notebook + data sample + report*

Web Scraping and APIs

Module II · Released Week 6

Objective	Collect digital trace data legally and reproducibly, by scraping a website or by querying a platform's API.
Data source	**Your choice** — one of the two tracks below. Practice sites built for scraping: books.toscrape.com, quotes.toscrape.com. APIs covered in class: YouTube Data API v3, Bluesky / AT Protocol, GDELT DOC 2.0.
Outputs	Folder `assignment_3/`: `A3_notebook.ipynb`, `A3_data.csv` (the data, or a sample of up to 5,000 rows), `A3_report.pdf` (source, legality, data dictionary).

Choose your track

Track A · Web scraping

Scrape a simple website that permits it. Check its **robots.txt** and terms of service first.

Track B · Platform API (optional alternative)

Use a platform's official API (YouTube, Bluesky, GDELT, Reddit, Wikipedia, …). Check its terms, quotas, and authentication.

Tasks

01 **Choose a source.** Describe it, say why it interests a social scientist, and confirm that collection is permitted.

02 **Collect.** Gather at least a few hundred records. Handle pagination and rate limits, pause between requests, and keep API keys in environment variables.

03 **Clean and store.** Flatten the output into a table, document how you handled missing or inconsistent entries, and save it as CSV. Add a one-line-per-column data dictionary to the report.

04 **Ethics and legality.** In the report, add a short paragraph on terms of service, personal data, and what would have to change for published research (Freelon 2018; Jünger 2018).

Assignment 04 *13 points · Notebook + target coding + one-page report*

Machine Learning — Shared Task

Module III · Released Week 9

Objective	Predict whether a social-movement organisation is **right-wing**. SOMOS records ideologies, not a left–right position, so you build the target yourself.
Dataset	**SOMOS**: 12,753 Turkish social-movement organisations on Twitter/X, coded for type, ideology, scope, theme, inequality axis, and city (Kına et al. 2026; see the annotation manual). On the course GitHub page as `Datasets/somos/SOMOS-Dataset.xlsx`.
Target	`right_wing` (1 or 0), built from `ideology_primary`.
Outputs	Folder `assignment_4/`: `A4_notebook.ipynb`, `A4_target_coding.csv` (your ideology-to-target mapping), `A4_report.pdf` (one page).

Tasks

01 **Build the target.** Group the ideology categories into right-wing (1) and not right-wing (0), justify each choice, and save the mapping. Decide what to do with *Unclear* organisations and whether `ideology_secondary` plays a role.

02 **Choose predictors.** Clean and encode the other columns and decide which to use. Justify each, and avoid leakage: a variable used to build the target cannot be a predictor.

03 **Model.** Compare at least three models against a majority-class baseline, tuned with cross-validation on the training rows.

04 **Evaluate.** On the test rows, report F1 for the right-wing class and a confusion matrix. Which predictors matter, and how much do results change if you re-code your most contestable category?

05 **Report.** One page: target coding, predictors, models, results, and what this prediction does and does not tell us.

> ***Shared task***
>
> **Test set:** organisations with `id % 5 == 0`; train on the rest. **Ranking:** F1 for the right-wing class, with accuracy breaking ties.
>
> Since everyone builds their own target, each score is shown next to its coding. Higher scores are rewarded only under a defensible coding.

Assignment 05 *8 points · Notebook + Excel file + one-page PDF report*

LLMs as Annotators

Module IV · Released Week 11

Objective	Use LLMs as measurement instruments: label a corpus whose true labels you know, and find which **model × prompting strategy** combination reproduces them best.
Dataset	**Your choice**: any text corpus with a trustworthy, human-assigned binary label. Cite its source. Ideas: your own hand-coded data (from Assignment 3 or your thesis); TweetEval (hate, offensive, stance); Jigsaw Toxic Comments; Manifesto Project quasi-sentences; Comparative Agendas Project policy topics; protest-event news corpora.
Outputs	Folder `assignment_5/`: `A5_notebook.ipynb` (prompts, API calls, evaluation), `A5_annotations.xlsx` (sheets: annotations | metrics | prompts), `A5_report.pdf` (one page).

Minimum requirements

Requirement	Minimum	Notes
Observations	100	More is welcome, cost permitting
Positive cases	10	Class of interest must appear at least 10 times
Models	3	From at least two providers is preferred
Prompting strategies	3	Same three across all models
Combinations evaluated	9	3 models × 3 strategies, full factorial

Both a rare-event design (**10 positives in 1,000**) and a balanced one (**50 in 100**) are fine. Say which you have, since it affects which metrics are meaningful.

Tasks

01 **Define.** State the construct, write a short codebook definition of the positive class, and describe the corpus and how its ground truth was produced.

02 **Design.** Choose three models (record exact identifiers and settings; temperature 0 is recommended) and three prompting strategies, e.g. zero-shot, few-shot, chain-of-thought. Few-shot examples must not come from the evaluation set.

03 **Annotate.** Run all nine combinations. Force a machine-readable answer, cache responses, and keep API keys out of the repository.

04 **Evaluate.** For each combination, report precision, recall, and F1 for the positive class, plus a confusion matrix. Name the best combination and look at its errors.

Excel output

One file, `A5_annotations.xlsx`, with three sheets: **annotations** (one row per text, the gold label, and one column per combination, as below), **metrics** (one row per combination, best highlighted), and **prompts** (the exact prompt texts).

id	text	gold	GPT·ZS	GPT·FS	GPT·CoT	Claude·ZS	Claude·FS	Claude·CoT	Gemini·ZS	Gemini·FS	Gemini·CoT
001	"They should all be sent back where they …"	1	1	1	1	1	1	1	0	1	1
002	"Great turnout at the park clean-up today!"	0	0	0	0	0	0	0	0	0	0
003	"…"	…	…	…	…	…	…	…	…	…	…

ZS = zero-shot · FS = few-shot · CoT = chain-of-thought. Name the columns after the models and strategies you use.

One-page PDF report

Research question and data; models and strategies; a 3 × 3 table or heatmap of F1; the best combination and its errors; limitations (data leakage, reproducibility, cost).

Optional extension

Train a small supervised classifier on the labels of your best combination and compare it with the gold standard.

Assessment (8 points)

Component	Points
Design: construct, data, ground truth, prompts	2
Implementation: nine combinations, clean, reproducible code	2
Evaluation: correct metrics, Excel file complete and structured	2
Report: best combination, error analysis, limitations	2